

Thematic Structures of Deception: A Topic Modeling Analysis of Embedded Lies by Motivational Context

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Abstract

INTRODUCTION: Historically, deception research has focused almost exclusively on detection through behavioral cues, overlooking how underlying motivation shapes the actual semantic content of a lie. This study shifts the focus from detection to content by investigating how motivational intent influences the thematic composition of "embedded lies."

OBJECTIVES: This research aims to determine if approach-oriented (*Approach*) and avoidance-oriented (*Avoidance*) motivations produce distinct thematic structures, interpreted through the content taxonomy of DePaulo et al. (1996).

METHODS: Using an event-stratified *BERTopic* pipeline, 3,256 embedded lies from $N = 657$ narratives were clustered into semantic topics. The pipeline was optimized using the *Optuna* framework to maximize a composite objective of topic coherence and diversity. These computational clusters were then mapped to a five-category taxonomy (*Feelings*, *Actions*, *Achievements*, *Explanations*, *Facts*) to identify the narrative functions of the fabrications.

RESULTS: The analysis revealed that approach-oriented lies prioritize *Achievements* and *Facts* to proactively construct desirable realities or enhance social status. In contrast, avoidance-oriented lies rely significantly on *Actions* for logistical denial and *Explanations* to provide external justifications and evade negative outcomes.

CONCLUSION: Deceptive strategies are not random fabrications; they are functionally and strategically driven by the deceiver's motivational context. This suggests that deception modeling should integrate the strategic "why" to better understand the semantic "what," providing a foundation for more sophisticated, motivation-aware detection systems.

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1 Introduction

Deception is a routine part of everyday communication, yet much of the existing literature has primarily focused on how deception can be detected. Research on behavioral and linguistic cues has shown that such indicators are highly context-dependent, suggesting that deception does not reflect a single, stable psychological state (Depaulo et al., 2003; Hauch et al., 2015). Increasingly, deception is therefore understood as a strategic act: what people choose to fabricate depends on their goals, constraints, and the situational context in which the lie occurs (Nahari, 2019).

This thesis shifts the focus from detection to content. Rather than asking whether a statement is deceptive, it examines *what* is being lied about and how that content varies as a function of motivational intent. The empirical focus lies on *embedded lies*: short, plausible falsehoods embedded within otherwise truthful narratives. Because such lies fit naturally into otherwise truthful stories, they provide a realistic view of how deception is used to achieve goals rather than being freely invented (Loconte and Kleinberg, 2025).

Embedded lies are analyzed under two motivational contexts derived from regulatory focus theory: *Approach* and *Avoidance*. Approach-oriented deception serves promotion-focused goals, such as gaining status or resources, whereas avoidance-oriented deception serves prevention-focused goals, such as evading blame or negative consequences (Crowe and Higgins, 1997; Elliot and Church, 1997). These differing goals are expected to yield systematic differences in semantic content. Approach lies should emphasize constructive claims about competence and outcomes, while avoidance lies should rely more on denial, justification, and procedural explanations (Nahari, 2019; Vrij, 2008).

To interpret these patterns, this study adopts the content taxonomy of DePaulo et al. (1996), which distinguishes lies concerning *feelings*, *actions*, *achievements*, *explanations*, and *facts* (DePaulo et al., 1996). This framework enables the translation of computationally derived topics into psychologically relevant categories.

Methodologically, the thesis applies an event-stratified *BERTopic* pipeline to the Embedded Lies dataset (Loconte and Kleinberg, 2025; Grootendorst, 2022). By combining transformer-based embeddings with density-based clustering, this approach captures semantic structure in short texts where classical topic models often fail (Devlin et al., 2019). The resulting topics are subsequently mapped to the DePaulo taxonomy, enabling a systematic comparison between approach- and avoidance-motivated deception.

2 Related Work

2.1 Psychological and Linguistic Perspectives

Meta-analyses show that non-verbal and physiological cues are unreliable indicators of deceit, as they often reflect stress rather than deception (Nahari, 2019; Depaulo et al., 2003). Linguistic tools like LIWC also show small, context-dependent effects (Pennebaker et al., 2001; Newman et al., 2003). Consequently, modern theory treats deception as strategic communication shaped by intent (Nahari, 2019). Embedded lies, which mix truthful and false information, illustrate this strategy by preserving narrative coherence, making them difficult to detect through surface-level markers (Loconte and Kleinberg, 2025).

2.2 Motivational Orientations

Regulatory Focus Theory distinguishes between approach and avoidance orientations (Crowe and Higgins, 1997). Approach orientation serves promotion goals like status or resource attainment. Avoidance orientation serves prevention goals, such as evading sanctions or social friction. These motivations likely trigger distinct semantic structures: one constructing a desired reality, the other protecting against negative consequences (Elliot and Church, 1997; Nahari, 2019).

2.3 Topic Modeling and BERTopic

Topic modeling identifies latent themes through word co-occurrence (Kherwa and Bansal, 2019). While models like LDA are standard for long-form text (Deerwester et al., 2015; Campbell et al., 2015), they struggle with short deceptive fragments due to data sparsity (Kherwa and Bansal, 2019; Zhu and Liu, 2025). Modern neural approaches using BERT generate context-sensitive embeddings that capture semantic nuances more effectively (Devlin et al., 2019; Ethayarajh, 2019).

This study utilizes BERTopic, which combines sentence transformers with UMAP for dimensionality reduction and HDBSCAN for density-based clustering (Grootendorst, 2022; McInnes et al., 2020, 2017). This pipeline allows for analysis of embedded lies by identifying regions of high semantic similarity (Zhu and Liu, 2025). Topic representations are refined using class-based TF-IDF (c-TF-IDF) and Maximal Marginal Relevance (MMR) to ensure thematic distinctness and relevance (Grootendorst, 2022; Carbonell and Goldstein, 1998).

2.4 Content-Based Taxonomy

To interpret deceptive content, this study applies the taxonomy of DePaulo et al. (1996), which identifies five primary categories of lies. First, *feelings* involve fabricated emotions, such as feigning enthusiasm during an interview or exaggerating pain for insurance compensation. Second, *actions* concern false whereabouts or deeds, like claiming to be at the office while at a bar. Third, *achievements* encompass self-enhancement, such as inflating professional accomplishments or claiming non-existent technical skills. Fourth, *explanations* are fabricated justifications, such as blaming a technical glitch or a personal emergency for a missed deadline. Finally, *facts* involve objective falsehoods regarding material details, such as misrepresenting an item’s cost or the specific details of a car accident. This framework enables a systematic mapping of semantic themes to the deceiver’s underlying motivation.

Grounded in ordinary interactions, this framework provides a standardized lens through which the semantic output of deceptive communication can be interpreted across motivational contexts.

3 Research questions

Based on the identified gap regarding the semantic content of deception, this study investigates the relationship between motivational intent and narrative structure.

Moving beyond simple classification, the Approach versus Avoidance framework is used to hypothesize that opposing goals lead to divergent thematic structures. To capture these nuances in short text, *BERTopic* is employed to map the resulting clusters to the taxonomy of DePaulo et al. (1996). This framework was selected for its strong alignment with the social nature of the *Embedded Lies* dataset Loconte and Kleinberg (2025).

3.1 Main Research Question

Specifically, the research seeks to address the following question:

MRQ: How do the core semantic topics of *embedded lies* differ when mapped to the content taxonomy of DePaulo et al. (1996), specifically regarding the distinction between approach-oriented (*Approach*) and avoidance-oriented (*Avoidance*) motivations?

3.2 Sub-questions

To provide a detailed analysis of this relationship, the main question is divided into three specific sub-questions:

- **SQ1:** Which specific content categories (e.g., *Achievements*, *Facts*) from the DePaulo framework characterize the themes of lies told for *Approach* (approach motivation)? We hypothesize that these narratives will exhibit features of reality construction and self-enhancement.
- **SQ2:** Which specific content categories (e.g., *Actions*, *Explanations*) from the DePaulo framework characterize the themes of lies told to avoid *Avoidances* (avoidance motivation)? We hypothesize that these narratives will rely on denial strategies and external justifications.
- **SQ3:** To what extent do the identified computational clusters align with the DePaulo taxonomy to reveal consistent ‘lying strategies’ across diverse narrative events? This question assesses the robustness of the taxonomic mapping across varying narrative contexts.

4 Methodology and analysis plan

This chapter outlines the methodological framework used to operationalize the relationship between deceptive motivation and semantic content. To address the research questions, we employ a mixed-methods design that bridges computational linguistics with psychological theory.

The analysis begins with the *Embedded Lies* dataset, described in **Section 4.1**, which serves as the source of naturalistic deceptive fragments. To isolate the specific impact of motivation on narrative structure, **Section 4.2** details the research design, including the partitioning of data into *Approach* and *Avoidance* contexts and the event-stratified modeling approach. A comprehensive schematic overview of this full research design is provided in **Appendix C**.

The computational core of the study is presented in **Section 4.3**, which describes the modular *BERTopic* pipeline used to extract topics, optimized via the custom objective metrics defined in **Section 4.4**. Finally, to bridge the gap between computational clusters and psychological theory, **Section 4.5** details the human-in-the-loop annotation protocol used to validly map these themes to the taxonomy of DePaulo et al. (1996).

4.1 Dataset overview

The analysis draws on the *Embedded Lies* dataset Loconte and Kleinberg (2025), a large-scale corpus specifically designed to study deceptive fragments within naturalistic narratives.

4.1.1 Data collection procedure

The data from the Loconte and Kleinberg (2025) was collected through a crowd-sourcing experiment involving $N_{\text{part}} = 1041$ participants. In the experimental setup, participants were presented with a series of everyday scenarios (e.g., job interviews, insurance claims) and instructed to write narratives based on these prompts. The protocol utilized a truth-default modification design: participants alternated between writing truthful stories based on genuine personal memories and deceptive stories where they fabricated key details. Crucially, for each event, participants were assigned a specific deceptive goal (see Table 4).

The dataset relies on self-annotation. After writing a deceptive narrative, participants were required to highlight specific spans of text that contained the core fabrication. These highlighted spans constitute the “embedded lies.” The complete raw dataset comprises $N_{\text{total}} = 5245$ embedded lies distributed across $N_{\text{events}} = 12$ distinct narrative contexts.

4.1.2 Data structure

The data possesses a hierarchical structure: a single participant provides multiple narratives, and crucially, a single narrative may contain multiple distinct deceptive fragments. Table 1 details the attributes in the original data set. While the raw data organizes these fragments as nested lists within a narrative object (wide format), the primary unit of analysis for this study is the individual *embedded lie text*.

Table 1: Attributes of the Embedded Lies dataset utilized in the analysis.

Attribute	Description
Embedded lie text	The specific deceptive fragment (short text) used as the primary input.
Event type	The narrative context in which the lie occurred (e.g., ‘Job Interview’).
Motivation	Annotation describing the reason for deception (stratified into Approach/Avoidance).
Source	Indicates the cognitive origin (e.g., personal experience vs. pure imagination).
Centrality	A numerical rating reflecting how central the lie is to the broader story.
Deceptiveness	The perceived intensity or magnitude of the lie.
IDs	Participant and event identifiers linking the fragment to the full narrative.
Length	The token or word count, used to guide preprocessing exclusion steps.

4.1.3 Preprocessing and filtering

Preprocessing starts with the raw dataset as provided by the Loconte and Kleinberg (2025) paper. To prepare this data for robust topic modeling, a technical preprocessing pipeline was applied.

First, the dataset was filtered to align with the research design defined in Section 4.2. This step removed narratives irrelevant to the target motivational contexts, reducing the dataset to $N_{\text{filtered}} = 657$ relevant narratives.

Second, the data was reshaped from a wide format (one row per narrative) to a long format. This involved “exploding” the nested columns so that each individual embedded lie constituted a single, independent row in the dataframe.

The final preprocessed dataset consists of $N_{\text{lies}} = 3256$ individual embedded lies across $N_{\text{events}} = 9$ narrative contexts, serving as the input for the topic modeling pipeline. An overview of the dataset can be found in 2

Table 2: Attributes of the preprocessed dataset used as input for the event-stratified BERTopic pipeline.

Attribute	Description
embedded_lie	The cleaned embedded lie text (unit of analysis) used as input for embeddings and topic modeling.
Event	Narrative event context used for event-stratified modeling (e.g., <i>A job interview for your dream job</i>).
is_positive_gain	Binary indicator of motivational context derived from event goals (1 = <i>Approach</i> , 0 = <i>Avoidance</i>).
Participant_id	Unique participant identifier linking multiple narratives from the same individual.
embedded_lie_n	Embedded lie identifier within a narrative (e.g., embedded_lie_4, is the 5th embedded lie in its scenario).
False_event	The full deceptive narrative text from which the embedded lie span was extracted (kept for traceability and qualitative checks).
False_event_num_words	Word count of the full deceptive narrative (used for descriptive statistics and sanity checks).
EL_num_words	Word count of the embedded lie fragment (used for preprocessing diagnostics and short-text sensitivity checks).

4.2 Research design: Data selection and stratification

Prior to computational analysis, the dataset underwent a curation and partitioning process. This ensured a balanced comparison between the two primary motivational contexts and removed confounding variables that could skew the semantic analysis.

4.2.1 Motivational partitioning and exclusion criteria

The core analytical distinction in this study is applying the Regulatory Focus Theory Crowe and Higgins (1997) established in Section 2.2. Specifically, we partition deceptive narratives into two distinct motivational types; *Approach* (aligning with promotion focus) and *Avoidance* (aligning with prevention focus). This classification is derived by mapping the specific deceptive goal of each narrative event (detailed in Table 4) to the operational definitions outlined in Table 3.

To ensure thematic cohesion and methodological rigor, specific exclusion criteria were applied to the raw dataset prior to analysis. Two romance-focused events, “Cheating on your partner” and “Ending a long romantic relationship”, were explicitly excluded. This decision was driven primarily by the need for thematic cohesion; these emotional, interpersonal scenarios differed from the functional, transactional nature of the other events (e.g., interacting with authority figures), and their inclusion would likely introduce distinct emotional vocabularies acting as noise in the semantic space. Furthermore, this exclusion

improved class balance. The raw dataset was heavily skewed towards avoidance narratives; removing these events reduced the *Avoidance* group size from 568 to 380, creating a more balanced comparison against the *Approach* group ($N_{\text{approach}} = 277$).

Additionally, the unclassified category *none of them* was removed as it lacked a defined motivational goal and thus cannot be mapped to neither avoidance or approach. The final dataset consists of 657 narratives distributed across nine distinct events, as detailed in Table 4.

Table 3: Operational definitions of motivational contexts used for dataset partitioning distilled from Crowe and Higgins (1997).

Motivational Context	Operational Definition
Approach	Lies told with the primary goal of acquiring a direct benefit, resource, or status enhancement. Examples include securing employment, obtaining financial compensation, or inflating social status.
Avoidance	Lies told with the primary goal of avoiding a punishment, or loss. Examples include escaping blame for an error, evading a financial penalty, or preventing professional repercussions.

4.2.2 Event-stratified modeling

A critical challenge in analyzing deceptive narratives is the semantic heterogeneity present across different contexts. A lie told during a professional job interview is likely to utilize a vastly different vocabulary than a lie told to explain a traffic accident. Applying a single, global topic model to such a diverse corpus would likely result in clusters defined by domain-specific nouns (e.g., “car”, “office”, “ticket”) rather than the underlying strategic content of the deception.

To mitigate this risk and preserve the event-level semantic structure, we adopted an event-stratified modeling approach. This design entails finetuning independent topic models for each distinct narrative event type. By doing so, the algorithm can detect subtle thematic variations within a specific context without being overwhelmed by cross-context vocabulary differences.

4.3 Topic modeling pipeline

This study employed *BERTopic* Grootendorst (2022), a modular topic modeling framework that leverages transformer-based embeddings and class-based TF-IDF Grootendorst (2022) to generate coherent topic representations. Unlike monolithic probabilistic models such as LDA, *BERTopic* allows individual stages of the topic modeling pipeline to be configured and optimized independently. This modularity is particularly important when modeling short and heterogeneous text fragments, where topic quality is highly sensitive to hyperparameter choices.

To systematically identify robust parameter configurations and avoid ad-hoc tuning, a dedicated hyperparameter optimization stage was implemented using the *Optuna* framework Akiba et al. (2019). In this research context, *Optuna* formulates hyperparameter selection as a black-box optimization problem, iteratively sampling and evaluating combinations of parameters governing UMAP, HDBSCAN, vectorization, and topic representation. Each configuration was evaluated using a custom objective function that

Table 4: Distribution of narrative events across motivational categories ($N = 657$), adapted from Loconte and Kleinberg (2025).

Event Context	N	Deceptive Goal
<i>Approach</i>		
A job interview for your dream job	160	Inflate your past experiences to get the job
Being hospitalised and undergoing surgery	70	Exaggerate some side effects to receive extra compensation from your health insurance
Being involved in a car accident	47	Increase the claimed amount of damage you received to get more money
Total Approach	277	
<i>Avoidance</i>		
Causing a car accident	15	Describe the event so that it's not your fault
Cheating on an exam	48	Describing how you passed the exam, given that you cannot admit that you cheated
Getting a speeding fine	62	Pretend it wasn't you driving the car that day
Getting fired	34	Pretend that you just had a bad day at work
Missing a deadline at work	97	Find excuses that allow you not to appear forgetful or disorganised
Taking public transport without a ticket	124	Convince the ticket inspector that they shouldn't fine you for not having the ticket
Total Avoidance	380	

combines embedding-based topic coherence, lexical topic diversity, and a penalty for unstable topic solutions. This optimization procedure was executed prior to training the final models, ensuring stability and reproducibility across all event-stratified topic modeling runs.

The BERTopic workflow is visualized in Figure 1.

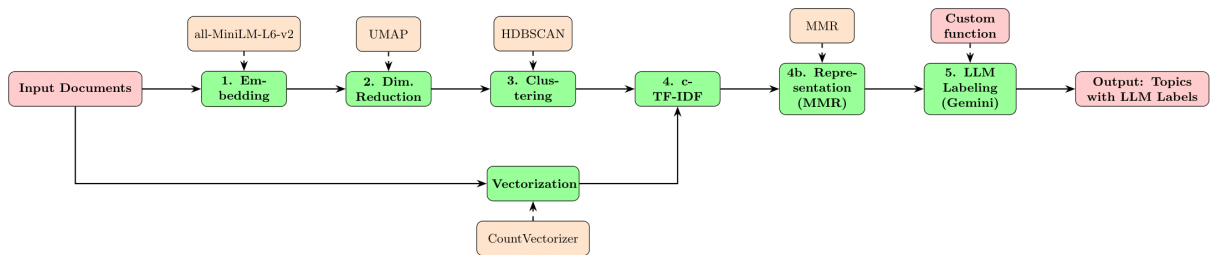


Figure 1: Schematic overview of the modular BERTopic pipeline, from sentence embedding to generative label assignment.

4.3.1 Embeddings

The first stage of the pipeline involves transforming the preprocessed text fragments into dense, high-dimensional vector representations. For this task, the `all-MiniLM-L6-v2` sentence-transformer model was employed. This model is built upon the Sentence-BERT framework Reimers and Gurevych (2019) and utilizes the MiniLM architecture Wang et al. (2020), which leverages deep self-attention distillation to achieve high computational efficiency without sacrificing semantic accuracy. This balance is particularly crucial for short-form text, where preserving subtle latent meanings is essential for identifying deceptive structures Loconte and Kleinberg (2025). To convert the token-level outputs into a single, fixed-size document embedding, a default mean-pooling strategy was applied. This approach ensures that the semantic essence of each fragment is captured within a continuous vector space, providing a robust foundation for subsequent clustering and thematic extraction Grootendorst (2022).

4.3.2 Hyperparameter optimization

Applying density-based clustering to short, variable-length texts is highly sensitive to parameter configurations. A generic configuration often results in either excessive noise (where too many documents are discarded as outliers) or over-generalized clusters. To address this, the model configurations were optimized independently for each of the nine narrative event subsets using *Optuna*.

For each subset, 25 trials were conducted using a Median Pruner (warmup steps=5) to maximize a custom objective function (defined in Section 4.4). To ensure reproducibility, a fixed random seed (`random_state=42`) was applied to all stochastic components, specifically the UMAP dimensionality reduction and sampling procedures. The search space, detailed in Table 5, was carefully designed to balance the preservation of local semantic structures with the formation of dense, interpretable clusters.

Table 5: Hyperparameter search space and fixed model settings used in the Optuna optimization.

Component	Parameter	Search Space / Setting
UMAP	Neighbors (<i>n_neighbors</i>)	Integer [5, 20] (step=5)
	Components (<i>n_components</i>)	Categorical {5, 10}
	Min. Distance (<i>min_dist</i>)	Float [0.0, 0.1]
	Metric	Cosine similarity
HDBSCAN	Min. Cluster Size	Integer [5, 40]
	Min. Samples	Integer [1, 15]
	Metric	Euclidean
	Selection Method	Leaf clustering (<i>leaf</i>)
Vectorizer	Min. Document Freq. (<i>min_df</i>)	Integer [1, 3]
	N-gram range	(1, 2)
	Stop words	Standard English list

4.3.3 Clustering and outlier handling

In contrast to algorithms like K-Means which force every data point into a cluster, this analysis utilized HDBSCAN, a density-based clustering algorithm. HDBSCAN is particularly suited for natural language data as it identifies "noise" points; outliers or embedded lies that are semantically distinct and do not fit well into any dominant theme. While this improves the purity of the resulting topics, it initially creates a "*Topic(-1)*" containing all documents classified as noise. To maximize data retention without compromising topic coherence, a post-hoc outlier reduction strategy was applied. Noise points were reassigned to their nearest topic cluster, provided the cosine distance remained below a strict threshold of 0.15. Documents exceeding this distance remained classified as Topic -1 and were subsequently excluded from the final dataset. This refinement resulted in a reduction from $N_{before} = 3,256$ to $N_{after} = 3,241$, representing the removal of 15 persistent noise points.

The extent of this reduction and its implications for data integrity are further addressed in the discussion 6.2.

4.3.4 Topic Representation and Labeling

Following the formation of clusters, topics were represented using Class-based TF-IDF (*c-TF-IDF*), an adaptation of the TF-IDF metric designed for document aggregates. To enhance interpretability, MMR was integrated into the extraction phase to minimize redundancy and promote lexical diversity within the keyword lists.

The labeling process utilized a semi-automated approach to facilitate consistency and rapid iteration. The primary keyword sets for each topic were processed by the `gemma-3-27b-it` model Team and DeepMind (2025) via the Generative AI API Google (2025). This model was prompted to generate concise labels of fewer than five words, specifically instructed to disregard noise terms. This automated labeling served as an essential tool for rapid qualitative assessment during the model refinement stages, allowing for immediate verification of topic coherence. To maintain rigorous standards, all generated labels underwent a final manual vetting process by the researcher to ensure precise alignment with the underlying semantic content. The complete prompt used is detailed in Appendix A.

The full codebase used in this research is available at: https://github.com/Wobknaap/BEP_1490621/tree/main

4.4 Evaluation metrics and optimization objective

To optimize the BERTopic pipeline, a composite objective function S_{final} was defined combining topic coherence (C_v) and topic diversity (D), with an additional penalty for incoherent (undefined) topics. This reflects the well-known trade-off in topic modeling between semantic coherence and topical distinctiveness Chagnon et al. (2024); Farea et al. (2024). Recent evaluations of BERTopic explicitly frame coherence and diversity as competing objectives that should be jointly optimized Kandala and Hoemann (2025), consistent with broader benchmarking observations for neural topic models Chagnon et al. (2024); Kherwa and Bansal (2019).

4.4.1 Optimzation fuction

Topic Coherence (C_v): Topic coherence was computed using an embedding-based metric. For each topic, coherence was derived from the semantic similarity between its top words using cosine similarity of their embedding representations. Topics containing fewer than two valid words were assigned an undefined coherence score and excluded from aggregation.

The overall coherence score was computed as the mean coherence over all valid topics:

$$C_v = \frac{1}{|T_{\text{valid}}|} \sum_{t \in T_{\text{valid}}} C_v(t), \quad (1)$$

where $C_v(t)$ denotes the embedding-based coherence score of topic t , following standard practice in BERTopic-style evaluations Grootendorst (2022); Chagnon et al. (2024).

2. Topic Diversity (D): Topic diversity was computed over the same set of valid topics as the ratio of unique words to total words across topic representations:

$$D = \frac{|W_{\text{unique}}|}{|W_{\text{total}}|}, \quad (2)$$

where W_{unique} is the set of unique words across all $\text{top_words}(t)$ for $t \in T_{\text{valid}}$, and W_{total} is the total number of words across these topic word lists.

3. Penalty for incoherent topics (P_{junk}): To discourage solutions that yield many topics with undefined coherence, a penalty factor was applied based on the proportion of valid topics:

$$P_{\text{junk}} = \frac{|T_{\text{valid}}|}{|T|}. \quad (3)$$

4. Objective Function: The final optimization score is defined as a weighted combination of coherence and diversity, scaled by the junk-topic penalty:

$$S_{\text{final}} = (\alpha \cdot C_v + \beta \cdot D) \cdot P_{\text{junk}}. \quad (4)$$

In this study, the weights were set to $\alpha = 4.0$ and $\beta = 6.0$. These values were determined empirically to reflect the analytical objective of prioritizing thematic distinctiveness over maximal semantic compactness. Emphasizing diversity reduces the risk of topic redundancy and prevents convergence toward a small number of dominant clusters. The inclusion of P_{valid} further discourages solutions that achieve high coherence or diversity by discarding large portions of the dataset .

The optimal hyperparameter configurations and their corresponding objective scores are reported in Appendix B (Table 8).

4.4.2 Model Synthesis and Data Consolidation

The final phase of the pipeline integrated the optimized components into a unified execution flow. For each of the nine narrative events, the production model was initialized using the optimal hyperparameter configurations identified during the Optuna search (see Appendix B). During this execution, the outlier reduction strategy was first applied to reassign

noise points based on the 0.15 cosine distance threshold. Subsequently, Maximal Marginal Relevance (MMR) was utilized to refine the topic representations, ensuring that the generated keyword lists were both semantically diverse and non-redundant.

These refined representations were then processed by the generative labeling module to assign human-interpretable titles to each cluster. In the final step, the results from all independent event pipelines were consolidated into a single master dataset. A post-hoc filter was applied to this collection to remove all remaining unassigned instances ($Topic - 1$), resulting in a filtered dataset of 29 verified topics. This aggregated output, containing the event metadata, representative documents, and thematic labels, serves as the definitive empirical basis for the taxonomic mapping and motivational analysis presented in the following sections.

4.5 Qualitative Analysis and Taxonomic Mapping

To interpret the 29 identified topics, a *human-in-the-loop* protocol was employed to map the computational clusters to the taxonomy of DePaulo et al. (1996). This phase represented the transition from automated pattern recognition to qualitative psychological interpretation, utilizing a structured three-step verification process: (1) label refinement, (2) coherence screening, and (3) functional mapping.

4.5.1 Step 1: Label Audit and Refinement

Prior to taxonomic mapping, the initial descriptive labels generated by the `gemma-3-27b-it` model were subjected to a manual audit. While the LLM provided an efficient high-level summary of the clusters, these labels were verified against the top keywords and representative documents to ensure precision. In cases where the AI-generated label was found to be too generic or misaligned with the semantic nuances of the cluster, it was manually refined to accurately reflect the underlying deceptive theme or be assigned as incoherent. This step ensured that the topical summaries remained strictly grounded in the raw textual data.

4.5.2 Step 2: Semantic Coherence and Noise Assessment

The second step involved an evaluation of the *semantic coherence* of each cluster. Topics were retained for analysis only if the representative documents displayed a unified and recognizable theme. If a cluster appeared to lack any internal consistency, it was classified as *incoherent*. This verification acted as a qualitative filter, ensuring that subsequent mapping was performed only on robust and interpretable topics.

4.5.3 Step 3: Functional Taxonomic Mapping and Stratification

In the final step, each coherent topic was analyzed to determine its alignment with the five DePaulo categories: *Feelings*, *Actions*, *Achievements*, *Explanations*, and *Facts*. To ensure validity, a cross-referencing strategy was utilized involving three distinct attributes: the representative documents, the MMR-refined keywords, and the validated topic label.

Within this process, mapping was determined by the narrative function observed in the source text rather than isolated keywords. For example, the keyword “doctor” was only mapped to *Explanations* if the documents revealed it was used as a justification for a behavior (e.g., an excuse for being late). Topics that did not align with the existing taxonomy were designated as *uncategorized*. Finally, the annotated master list was

stratified by motivational category (*Approach* or *Avoidance*) to facilitate the extraction of motivation-specific thematic structures.

5 Results

This section reports the outputs of the event-stratified *BERTopic* pipeline and the subsequent mapping of topics to the DePaulo et al. (1996) content taxonomy. The analysis targets $N = 3,256$ embedded lies across *Approach* and *Avoidance* motivational contexts.

5.1 Topic modeling output and data coverage

Across the nine narrative events, the optimized *BERTopic* runs produced 29 distinct topics after consolidating event-level solutions.

Manual coherence screening indicated that 26 of 29 topics were semantically coherent and could be mapped to the DePaulo taxonomy, yielding a 89.66% mappable-topic rate. The complete topic list, including events, keywords, representative snippets, and assigned DePaulo categories, is reported in Table 6.

5.2 Thematic distribution by motivational context (SQ1 and SQ2)

To address SQ1 and SQ2, topics were assigned to one of the DePaulo content categories or labeled *Uncategorized* when no clear alignment was present. Table 7 summarizes the category distribution per motivational context, which is visualized in Figure 2.

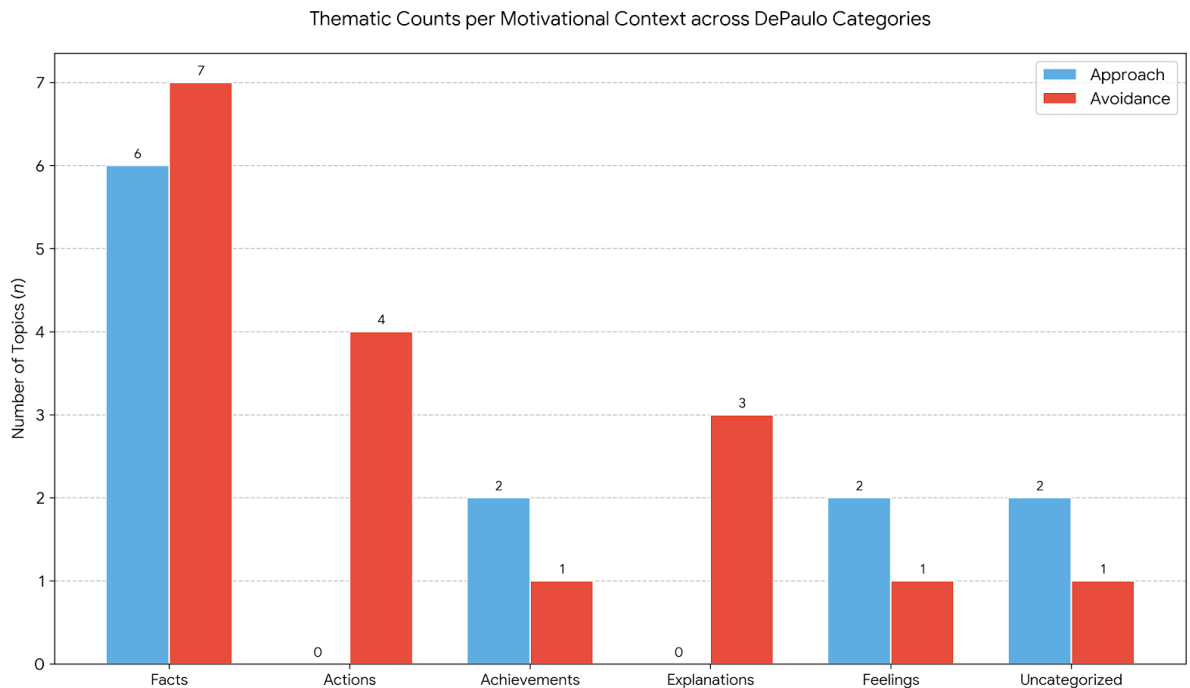


Figure 2: Distribution of DePaulo content categories across Approach and Avoidance motivations, illustrating the strategic shift from achievement-oriented to excuse-oriented deception.

5.2.1 Approach-oriented themes (SQ1)

Approach-motivated deception comprised $n = 12$ topics. The distribution concentrated in *Facts* ($n = 6$) and *Achievements* ($n = 2$). Substantively, job interview topics emphasized status-enhancing content such as "Career aspirations and experience" and "Technical questions" (Table 6), consistent with proactive goal pursuit. *Feelings* appeared in $n = 2$ topics and primarily reflected inflated confidence or amplified symptoms. Notably, *Actions* and *Explanations* were absent in the Approach set ($n = 0$ each), suggesting limited reliance on logistical alibis or justificatory excuse-making when the goal is gain acquisition rather than sanction avoidance.

5.2.2 Avoidance-oriented themes (SQ2)

Avoidance-motivated deception comprised $n = 17$ topics and exhibited a broader spread across the taxonomy. While *Facts* remained frequent ($n = 7$), the Avoidance condition was characterized by the presence of *Actions* ($n = 4$) and *Explanations* ($n = 3$). Representative snippets (Table 6) suggest that these categories functioned defensively: *Actions* often fabricated whereabouts or procedural behavior (for example, commuting narratives), whereas *Explanations* provided external justifications for negative outcomes such as missed deadlines or being late, frequently shifting blame to illness, organization, or circumstances.

5.3 Taxonomic Alignment and Potential Trends (SQ3 and MRQ)

Regarding SQ3, the successful mapping of computational clusters to the DePaulo et al. (1996) framework demonstrates a robust alignment between deceptive intent and thematic structure across diverse narrative events. The results indicate that the clusters are consistently interpretable within existing psychological categories, suggesting that the DePaulo taxonomy remains a stable lens for analyzing automated topics. While these findings should be viewed as contextual tendencies rather than definitive strategies, the systematic nature of the mapping supports the validity of the taxonomic approach.

With respect to the specific orientations, approach-oriented lies (SQ1) appear to lean toward *Facts* and *Achievements*, pointing to a focus on proactive reality construction. In contrast, the results for avoidance-oriented lies (SQ2) show a broader spread involving *Actions* and *Explanations*, suggesting a shift toward defensive management and the use of alibis. While the distributional contrast in Table 7 suggests a relationship between motivation and content, these patterns remain tentative and highly dependent on the narrative context. Consequently, the answer to the MRQ remains nuanced: motivational orientation is a relevant, but context-dependent, factor in shaping the thematic composition of an embedded lie.

5.4 Data Retention and outliers

The qualitative screening of the 29 topics identified specific instances of thematic fragmentation. As reported in Table 6, three topics (10.3%) were classified as *Incoherent* due to a lack of semantic unity in their top keywords and representative snippets.

Furthermore, clusters within the car accident scenarios were observed to be shorter and more descriptive than those in interview or workplace contexts. While these clusters remained interpretable as *Facts*, they contained a higher density of objective situational descriptions (e.g., "truck overturned," "hit") with fewer terms reflecting strategic construction.

Table 6: All 29 topics grouped by Motivation and Event, with keywords, representative snippet, and DePaulo category.

Motivation	Event	ID	Topic Label	MMR Keywords	Representative Snippet	DePaulo
Approach-Oriented Topics						
Approach	A job interview for your dream job	0	Job interview process	interview, interviewer, went, panel, going	and the interview went very well	Facts
		1	Incoherent	company, years, events, team, experience	company is great	Uncat.
		2	Daily routines and time	time, minutes, morning, late, office	I even went for a celebratory cup of coffee and cake before returning to work	Facts
		3	Emotional states and self perception	confident, nervous, looked, felt, excited	confidence booming and feeling prepared	Feelings
		4	Job offer communication	called, position, offer, email, week	They called to inform me that I got the job	Facts
		5	Career aspirations and experience	working, dream, company, wanted, school	This opportunity came in my life at the right time...	Achiev.
		6	Technical questions	questions, performance, explain, responses, data	Developed a machine learning model that reduced data processing time by 30%	Achiev.
Approach	Being hospitalised and undergoing surgery	0	Incoherent	little, heart, money, rushed, sick	he is doing sports little by little as a hobby	Uncat.
		1	Complications and recovery	surgery, pain, infection, surgeon, cyst	s days passed and I was taking care of my surgical scars...	Facts
		2	Hospital Experience	hospital, room, help, waiting, nurses	the healthcare workers didn't pay any attention to me...	Facts
Approach	Being involved in a car accident	0	Vehicle Accidents & Traffic	car, road, accident, truck, damage	a truck overturned and we somehow got hit...	Facts
		1	Injury and recovery process	leg, hospital, pain, fault, circus	i had to get a regular treatment once a week in Brussels	Feelings
Avoidance-Oriented Topics						
Avoidance	Causing a car accident	0	Traffic accident description	car, brake, hit, taxi, causing	The man got out of his car and came to my window...	Facts
		1	Social Connection and Awareness	friends, paying, sleepy, attention, listen,	knowing that the next day is a Monday and I will be working	Facts
Avoidance	Cheating on an exam	0	Student Exam Preparation	exam, studied, notes, paper, prepared	So, the day before the exam I was revising the whole day for it...	Achiev.
		1	Incoherent	attending, don remember, participate, festival	covered	Uncat.
Avoidance	Getting a speeding fine	0	Social outings with car travel	driving, car, friend, wife, drinks	i wasn't driving the car	Facts
		1	Traffic Violations	speeding, ticket, speed, road, limit	he was the one who got the speeding fine	Facts
Avoidance	Getting fired	0	Employment terms and conditions	job, salary, contract, team, month	I was desperate so I hired a bunch of people to create chaos...	Facts

(continued on next page)

Motivation	Event	ID	Topic Label	MMR Keywords	Representative Snippet	DePaulo
Avoidance		1	Daily commute and work life	went, late, target, work, office	I told them I could not go, I was not in the right mind space...	Actions
		2	Workplace hierarchy	manager, access, HR, database, supervisor	what kind of data I access in their database	Facts
	Missing a deadline at work	0	Workplace tasks and direction	project, boss, colleague, instructions	So, at work I was given this project that I was responsible	Actions
		1	Timing	time, test, early, connection, hour	we tried to test even absurd use cases	Facts
		2	Workplace communication & deadlines	deadline, manager, writing, email, info	missing a deadline	Explan.
		3	Team dynamics	team, colleagues, bad organisation, members	When asked why I had a semi-skilled team instead of my normal team...	Feelings
	Taking the bus/train without ticket	4	Illness	hospital, kid, broke, fell, clinic	We get to the clinic and an orderly seats him on a wheelchair...	Explan.
		0	Public Transportation Ticketing	ticket, inspector, bus, card, said	What I had not anticipated was that in Augsburg the ticket inspectors...	Actions
		1	Being late	late, time, hospital, hurry, didn	because I had more emergencies at the hospital that day and I couldn't leave earlier	Explan.
		2	Public Transportation & Commuting	train, station, stops, city, got	from the train station to the city center	Actions

Table 7: Thematic counts (n) per motivational context across DePaulo content categories.

Motivation	Facts	Actions	Achiev.	Explan.	Feelings	Uncat.	Total
Approach (n)	6	0	2	0	2	2	12
Avoidance (n)	7	4	1	3	1	1	17
Total	13	4	3	3	3	3	29

6 Discussion

The findings of this study provide a computational validation of the functional nature of deception. By analyzing the thematic structures of embedded lies, we demonstrate that deceptive content is not merely a byproduct of imagination but a strategic response to motivational pressures, aligning with the "lying in everyday life" framework established by DePaulo et al. (1996).

6.1 Contextual Influence on Modeling and Classification

A core premise of this research is that lying is fundamentally a *contextual act*. As suggested by Nahari (2019), the linguistic composition of a lie is inextricably linked to the social and goal-oriented framework in which it occurs. To capture this, an event-stratified modeling approach was utilized as detailed in Section 4.2.2. By training independent models for

each event, we prevented domain-specific nouns from dominating the latent space, which is a known risk in global models Campbell et al. (2015); Deerwester et al. (2015).

However, this contextual dependency introduces challenges regarding the stability of deceptive taxonomies. The results indicate that the determination of a DePaulo label is highly sensitive to the specific motivational pressures of an event. Because deceptive intent is strategic, the thematic content, and thus the assigned label, is a function of the underlying goal. This introduces a degree of "contextual flexibility" where the representative documents used for topic modeling provide the specific lens that dictates the label.

This implies that the reliability of such labels is subject to variance similar to that observed in human annotation. Just as the human participants in the Loconte and Kleinberg (2025) research may annotate different snippets of lies, the model's output is contingent upon the nature and length of the representative documents. If the context were expanded, a deceptive fragment might shift between categories, such as moving from *Action* to *Fact*. Thus, the resulting taxonomies should be viewed as dynamic interpretations influenced by the document context rather than fixed categories.

Also, while embedding-based topic models are highly effective at capturing dominant semantic patterns across large document collections, their outputs remain inherently interpretative and sensitive to modeling choices Reimers and Gurevych (2019); Boyd-graber et al. (2009).

6.2 Methodological Robustness and Data Constraints

The use of an automated objective function via Optuna, as described in Section 4.4, allowed for a systematic exploration of the hyperparameter space for HDBSCAN McInnes et al. (2017) and UMAP McInnes et al. (2020). Nevertheless, a persistent challenge in unsupervised topic modeling remains the absence of a definitive ground truth Kherwa and Bansal (2019). The optimal semantic resolution remains a matter of theoretical alignment rather than pure mathematical certainty.

Furthermore, sample size constraints for specific events impact the reliability of density-based clusters. While HDBSCAN is robust to noise, small datasets limit the model's ability to identify diverse sub-themes. To mitigate the loss of semantically relevant data, a post-hoc noise reassignment strategy was implemented. Manual inspection of the initial outliers (classified as Topic -1) revealed that many fragments were semantically aligned with existing clusters but resided in lower-density regions of the vector space. By applying a cosine similarity threshold of 0.15, documents initially labeled as noise were reassigned to their most proximal cluster.

This procedure proved to be highly effective for data retention: out of the original $N = 3,256$ embedded lies, only 15 fragments (approximately 0.46%) remained classified as persistent noise and were subsequently removed. Although a K-means approach was initially considered to ensure that every data point was assigned to a cluster, it was ultimately rejected. Unlike K-means, which forces every document into a cluster regardless of its distance from the centroid, HDBSCAN preserves the underlying topological structure of the data. The chosen hybrid approach strikes a necessary balance: it prevents the pollution of thematic cores with unrelated noise while ensuring that the final analysis is based on the most comprehensive subset of the data possible.

6.3 Thematic and Textual Heterogeneity

A significant limitation concerns the heterogeneity of self-reported lies, which ranged from single-word snippets to full sentences. While transformer models are proficient at capturing semantic nuances Devlin et al. (2019), their effectiveness is reduced when input lacks sufficient syntactic structure. As noted by Ethayarajh (2019), the geometry of BERT embeddings is highly sensitive to context.

A short snippet contains less semantic information than a full-sentence fabrication, as it lacks broader contextual structure. If a participant provides the entire narrative surrounding a lie, the model might capture the strategic intent more effectively. The current treatment of each annotation as an independent unit may therefore introduce noise. Future work could benefit from an enrichment step to better capture how a lie is structurally anchored within the surrounding truthful text Depaulo et al. (2003).

6.4 Generative Labeling

The integration of generative models, such as Gemma 3 Team and DeepMind (2025), for topic labeling represents a significant shift toward automated summarization, a process discussed in Section 4.5. This approach has recently been adopted as a core feature within the BERTopic framework Grootendorst (2022) to enhance the interpretability of high-dimensional clusters. While these models are highly effective at identifying dominant shared meanings across documents, they introduce an inherent layer of subjectivity. The framing of the generative prompt inherently directs the attention of the model, which may mirror the confirmation biases typically observed in human labelers. Consequently, while automated labeling increases efficiency, the resulting themes must be understood as products of both the latent data structure and the specific instructions provided to the generative agent.

6.5 Generalizability and Motivational Stakes

The generalizability of these findings is further constrained by the stakes of the experimental design utilized in this study. In high-stakes environments, such as criminal interrogations involving potential imprisonment, the cognitive load of the deceiver increases significantly Newman et al. (2003). Research suggests that in such contexts, deceptive strategies often shift toward "prevention-focused" inclinations aimed at minimizing information to avoid contradiction and "giving it away" Vrij (2008). This contrasts with the "promotion-focused" elaborate justifications frequently observed in low-to-moderate stakes scenarios, where deceivers are more concerned with being persuasive than with avoiding detection through silence Depaulo et al. (2003). Future research should therefore investigate whether the thematic distribution of *Achievements* and *Explanations* remains consistent when the risk of detection carries more severe consequences.

6.6 Future Work

The semantic and motivational patterns identified in this study suggest several directions for future research. Building on Loconte and Kleinberg (2025), subsequent work should focus on hierarchical modeling to connect event-level variation with global deceptive strategies. A key technical goal is the automatic extraction and embedding of the text surrounding deceptive statements. Providing transformer models with this contextual window, rather than isolated snippets, may reduce the snippet-context dilemma and

improve modeling of how deception is embedded within narrative structure Ethayarajh (2019).

To address the methodological circularity inherent in the current ground truth, future studies should move from fully unsupervised approaches toward guided or seeded topic modeling. Using the DePaulo et al. (1996) categories as semantic priors would enable more principled evaluation of psychological themes beyond post-hoc interpretation.

Finally, the robustness of the identified promotion and prevention patterns should be tested in high-stakes forensic settings. Whether these strategies persist under conditions of severe legal or professional risk remains an open question and a necessary step toward a more generalizable model of human deception.

7 Conclusion

Historically, deception research has been dominated by the search for universal behavioral cues, often neglecting the strategic substance of the deceptive message itself. This study addressed this gap by shifting the analytical focus from *how* people lie to *what* they choose to fabricate, specifically examining how motivational contexts shape thematic content. By applying an event-stratified *BERTopic* pipeline to a corpus of *embedded lies*, we demonstrated that deceptive themes are not random imaginative constructs but are functional responses tailored to the deceiver’s goals.

The findings provide suggest support for the integration of Regulatory Focus Theory within deception modeling. Our analysis reveals a divergence in communicative strategy where approach-oriented lies, driven by a promotion focus, function primarily as tools for proactive reality construction. These narratives leverage *Achievements* and *Facts* to enhance social status or secure desired gains. In contrast, avoidance-oriented lies, driven by a prevention focus, serve as mechanisms of defensive management. These rely on *Explanations* and *Actions* to navigate logistical threats or evade sanctions.

Methodologically, this thesis demonstrates that event-stratified topic modeling, supported by automated hyperparameter optimization and human-in-the-loop taxonomic mapping, is a robust framework for capturing semantic nuances in short, fragmented deceptive fragments. While "Facts" and "Actions" form a common narrative baseline across most lies, the strategic shift in peripheral themes confirms that deception is a highly contextual and goal-driven act. In conclusion, this research moves beyond simple detection by providing a thematic blueprint of deceptive strategies. These insights try to offer a foundation for more sophisticated, motivation-aware detection systems and contribute to a deeper theoretical understanding of deception as a structured and functional instrument in human communication.

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A Topic Labeling Prompt

The following prompt was used to generate the labels:

You are an expert in topic modeling.
 You will receive a list of keywords that may contain noise or mixed subthemes.
 Your task is to generate one clear label (max. 5 words) that reflects the strongest common theme.

Rules:

- Choose the dominant shared meaning
- Ignore outliers
- Use human language
- One phrase, no punctuation

Now label:
 {keywords_str}

B Hyperparameter optimization results

Table 8: Optimal hyperparameters and best scores per event type. Scores are based on a weighted combination of topic coherence and diversity.

Event Type	UMAP			HDBSCAN		Vectorizer	MMR	Metric
	N_{nb}	N_{comp}	Min Dist	Min Clust	Min Samp	Min DF	Div	Score
A job interview for your dream job	30	10	0.07	27	10	1	0.67	7.20
Being hospitalised and undergoing surgery	25	10	0.06	12	10	1	0.52	7.14
Being involved in a car accident	20	5	0.05	25	3	1	0.20	7.35
Causing a car accident	10	10	0.04	10	7	1	0.56	7.11
Cheating on an exam	15	10	0.06	40	4	2	0.71	7.18
Getting a speeding fine	25	10	0.02	50	10	2	0.80	7.08
Getting fired	25	10	0.07	12	10	1	0.62	7.23
Missing a deadline at work...	15	5	0.01	33	10	2	0.60	7.15
Taking the bus/train without the ticket	10	5	0.10	42	10	2	0.78	7.20

C Research Design Overview

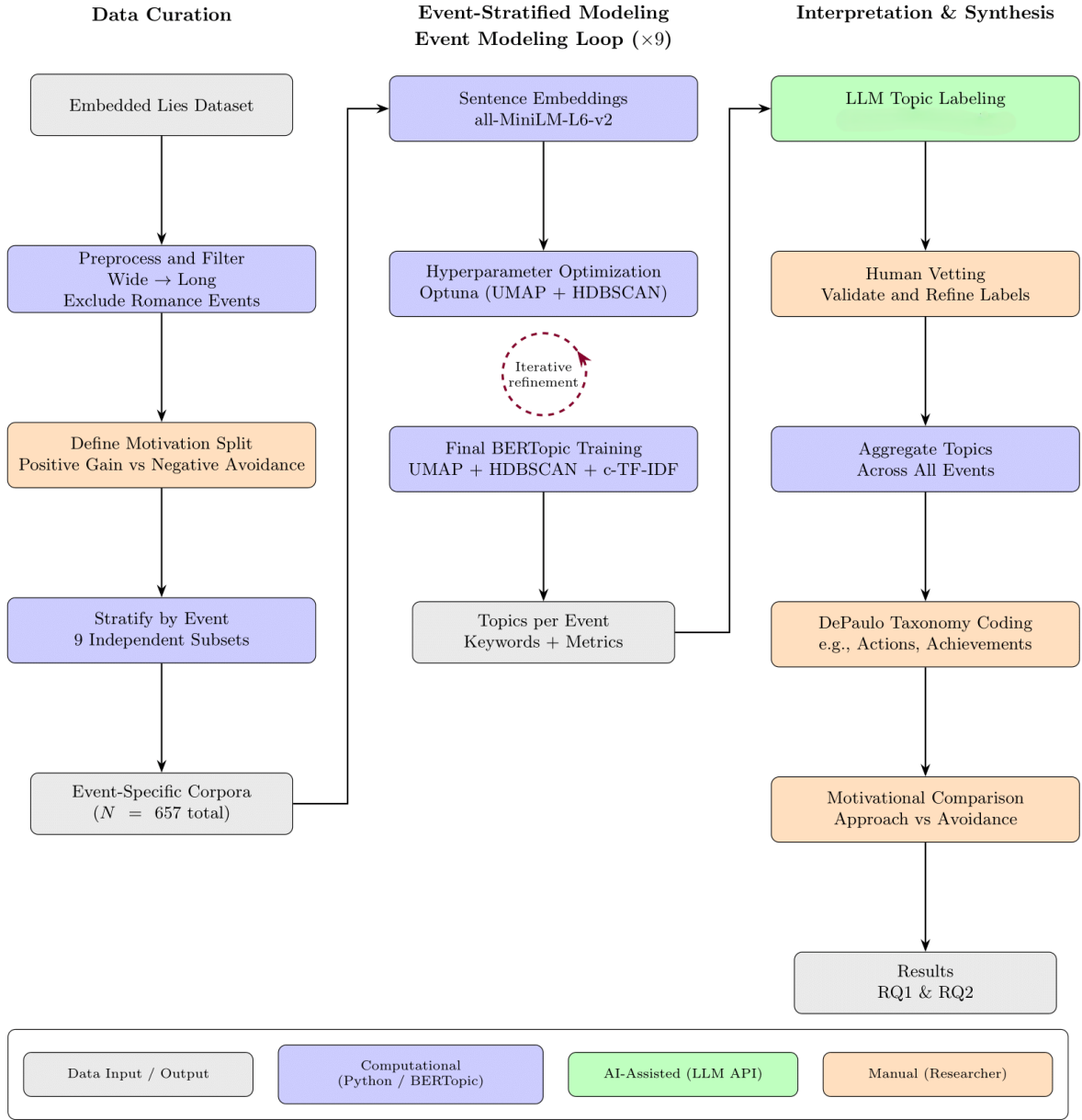


Figure 3: High-level overview of the event-stratified research design. The modeling loop is executed independently for each of the nine narrative events, allowing hyperparameters and clustering structure to adapt to event-specific semantic contexts.

Motivation	Full Event Name	Topic Label	MMR words	Key-	Representative Snippet	DePaulo
Approach	A job interview for your dream job	Technical questions	questions, performance, data		"Developed a machine learning model that reduced data processing time by 30%"	Achiev.
...						
Approach	Being hospitalised and undergoing surgery	Incoherent	little, heart, money, sick		"he is doing sports little by little as a hobby"	Uncat.
...						
Approach	Being involved in a car accident	Vehicle Traffic	car, road, accident, truck		"a truck overturned and we somehow got hit..."	Facts
...						
Avoidance	Taking the bus/train without ticket	Being late	late, time, hospital, hurry		"because I had more emergencies at the hospital that day..."	Explan.
...						
Avoidance	Taking the bus/train without ticket	Ticketing	ticket, inspector, bus, card		"...in Augsburg the ticket inspectors..."	Actions